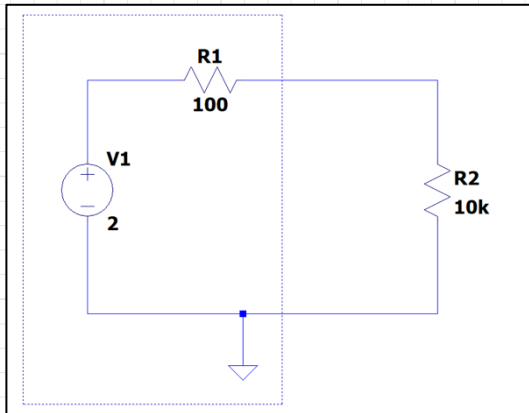
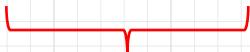


Lab 1: Basic principles

Task 1: output resistance





 Voltage Source with internal resistance Load resistance

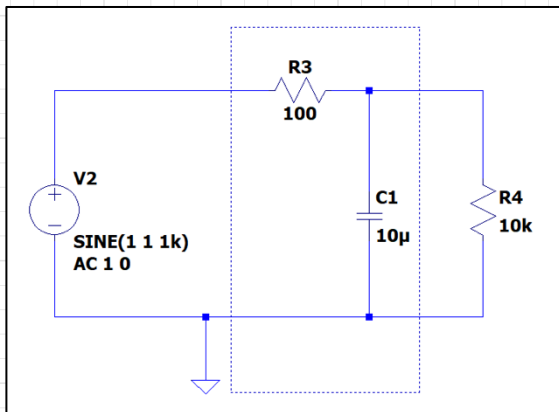
The first task is to explore the effect of output resistance. Use LTSpice to create the circuit to the left.

Set: $R2 = 10\text{k}\Omega$, $V1 = 2\text{V DC}$

1. What happens to the current through the load ($R2$) as $R1$ varies between 100Ω and $100\text{k}\Omega$?
2. What happens to the voltage across the load as $R1$ varies between 100Ω and $100\text{k}\Omega$?

The easiest way to determine this is to run repeated transient simulations. Ps. Look up the “potential divider”!

Task 2: Low pass filter






 Ideal voltage source RC Filter Load

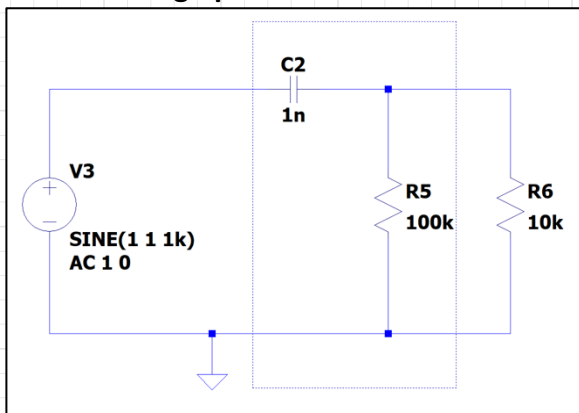
Now lets look at a low pass filter.

Set $R3 = 100\Omega$, $R4 = 10\text{k}\Omega$,

Set $V1 = 2\text{V AC @1kHz}$

1. Do a transient analysis and explore what happens to the sine wave amplitude as $C1$ varies from 100nF to $100\mu\text{F}$. Describe what is happening!
2. Do an AC analysis with $C1 = 1\mu\text{F}$. Determine the cut-off frequency from the bode plot.

Task 3: High pass filter






 Ideal voltage source RC Filter Load

Now lets look at a high pass filter.

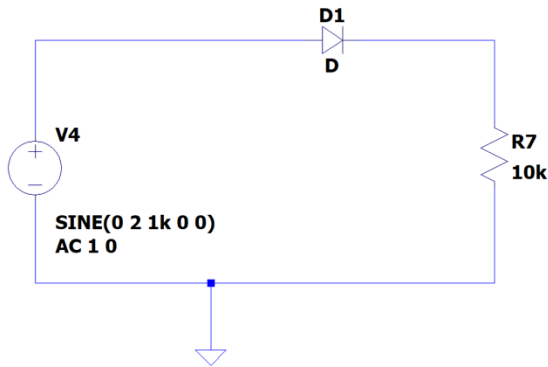
Set $R5 = 100\text{k}\Omega$, $R6 = 10\text{k}\Omega$,

Set $V1 = 2\text{V AC @1kHz}$

1. Do a transient analysis and explore what happens to the sine wave amplitude as $C2$ varies from 1nF to $1\mu\text{F}$. Describe what is happening!
2. Do an AC analysis with $C1 = 1\text{nF}$. Determine the cut-off frequency from the bode plot. Why is the answer different from what you would expect if you calculate the time constant as $\tau = 1/2\pi R_5 C_2$?

Lab 1: Basic principles

Task 4: Diodes

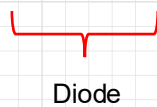


Now let's consider a diode circuit.

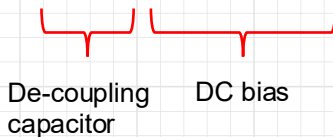
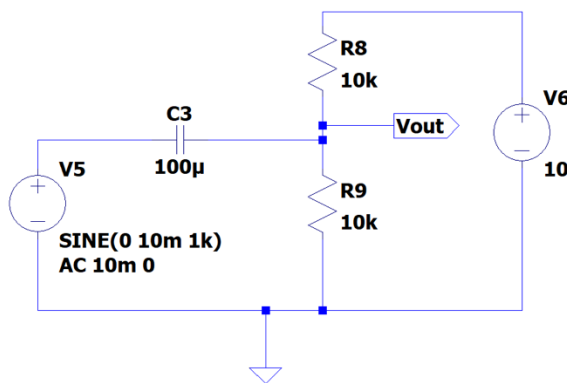
Set $V1 = 2V \text{ +/- AC @1kHz}$

Set $R7 = 10 \text{ k}\Omega$

1. Do a transient analysis and compare the voltage before and after the diode. What do you observe?
2. Change the value of $R7 = 1 \text{ k}\Omega, 5 \text{ k}\Omega, 10 \text{ k}\Omega$ and get the current through $R7$ in each case (using a transient simulation). Plot current vs $(1/R7)$, and determine the threshold voltage of $D1$.



Task 5: Small signal large signal



Set up a bias $V6 = 10V$, and put it across a potential divider consisting of $R8 = 10 \text{ k}\Omega, R9 = 10 \text{ k}\Omega$.

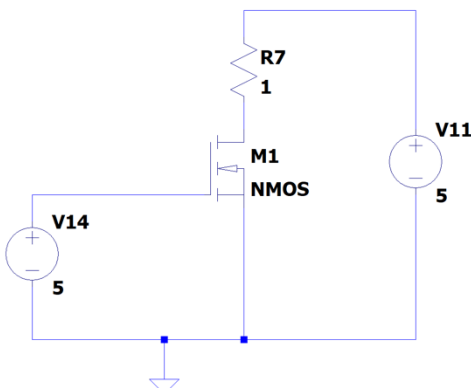
You can name the point between $R8$ and $R9$ as V_{OUT} .

Create a de-coupling capacitor $C3 = 100\mu F$.

Set $V5 = 10mV \text{ AC @1kHz}$.

1. Do a transient analysis on the voltage at V_{OUT} . Describe the waveform. What is the value of the large signal (DC) and small signal (AC) components?
2. What is the purpose of $C3$?

Task 6: Analysis of an nMOS transistor



Place an nMOS transistor and place a 1Ω resistor above it (to help with simulations, it can also be a 0Ω resistor).

1. Do a DC scan while varying the gate voltage V_g from $0V$ to $5V$. Obtain the current vs V_{gs} ($=V_g$ in this case) curve. What is the threshold voltage?
2. Do a parametric DC scan: Scan the V_{DS} by varying $V11$ from $0V$ to $5V$, then set the 2nd source V_g ($V14$) from 0 to $5V$ in $1V$ steps. Obtain the resultant current through the resistor $R7$. Identify the saturation and triode regions.

Lab 1: Basic principles

Some additional things to ponder in your own time:

Task 1:

1. Imagine you are trying to record a small voltage from a heart through an electrode. What happens if the electrode impedance is higher than the input impedance of your recording circuit (load)?

Task 2:

2. If you want to record a signal with a frequency range of 1-20 Hz, how do you remove high frequency noise?

Task 3:

3. If you want to separate a low frequency component from a signal, what would you do?

Task 4:

4. How do you remove the negative component of a signal or power supply?

Task 5:

5. How do you add a DC offset to an AC signal? (note, this is important later on!)

Task 6:

6. Although bipolar junction transistors (BJTs) are found in many discrete amplifiers, the microelectronics industry is now almost entirely using MOS Field effect transistors (MOSFETS).